No.	Status	Date
LNDE-RPT-20220906-033-DFR-1	Issued	2022-09-15

Daily Field Report 1 - UT Thickness Survey - ATU-100 Sour Gas, Regenerator Overhead, Rich Amine

Scope and Responsibilities

Marcus Chen, acting as lead UT technician, performed a UT thickness survey on five piping lines within the ATU-100. The scope of work was executed in accordance with the requirements of QSCP-NDT-001 Rev 2 and the safety guidelines established in LNDE-JSA-20220906-033.

All field activities were conducted under the authorized work permit QSCP-WP-2022-09151. Marcus Chen was responsible for the measurement of wall thickness and the verification of instrument calibration prior to data collection.

Equipment and Calibration

Thickness measurements were obtained using two Olympus 38DL PLUS units. The first unit (2847193) was verified against LNDE-CAL-2022-0791, and the second unit (2901456) was verified against LNDE-CAL-2021-0754.

Calibration was performed using LNDE-CS-001, a 4-step carbon steel calibration block ranging from 0.100 to 0.400 in. All measurement procedures adhered to the standards defined in ASTM E797.

Inspection of Sour Gas Inlet Lines

UT thickness measurements were recorded for 6-SG-101 and 6-SG-102. Both lines consist of A106 Gr.B material with an outer diameter of 168.3 mm and a nominal thickness of 7.11 mm.

The survey focused on verifying current wall thickness against the minimum retirement thickness of 3.56 mm for these sour gas inlet lines.

Inspection of Rich Amine and Regenerator Overhead Lines

The rich amine line 4-RA-101, constructed from A106 Gr.B with an outer diameter of 114.3 mm and nominal thickness of 8.56 mm, was surveyed against a minimum thickness of 3.81 mm.

Regenerator overhead lines 3-RV-101 (OD 88.9 mm, t_nom 7.62 mm, t-min 3.56 mm) and 2-RV-102 (OD 60.3 mm, t_nom 5.54 mm, t-min 2.79 mm) were also measured. Both lines are A106 Gr.B material.

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2022-09-15	_____